

Global architecture direction

The system should be built as a stack, not as disconnected setup scripts:

config

prepares machines, roles, users, mounts, package policy, Python environments, Runpod workspaces, publishing paths, and safe runtime checks

Agentfield

owns experiment intent, controller flow, agent selection, execution state, status, result tracking, and later campaign-level experiment automation

Paperclip-Agentfield adapter

connects Paperclip jobs/actions to Agentfield experiments and returns status, metrics, artifacts, and failure reasons

Paperclip

gives the human-facing dashboard, inbox, governance, job overview, review loop, and end-to-end visibility

That matches your Agentfield notes: config prepares users, environments, mounts, and packages; Agentfield becomes the experiment-aware layer using `GRNExperiment`, controller status/results, an agent registry, a reasoner invoker, and GRN agents. The Paperclip notes add the missing bridge: Paperclip should stay the UI/control surface, while an adapter translates Paperclip jobs into Agentfield runs and returns status/results.

Layer grouping

I would group the real-world work into **five layers**.

Layer 1 – Runtime substrate

1. Runpod portable runtime base
7. Remote model brain endpoint

Layer 2 – Role workstations

- 2. Research Scientist GRN workspace
- 6. AI Engineer agent/platform dev environment
- 9. Obsidian writing machine
- 10. Publisher LaTeX/paper export

Layer 3 – Research execution loops

- 3. NCA-ART research stack
- 4. Parameter search comparison tools
- 5. Runpod training/inference loop

Layer 4 – Knowledge, reasoning, and writing automation

- 8. OpenClaw + PKM reasoning workspace

Layer 5 – Platform orchestration

- 11. Agentfield experiment runtime foundation
- 12. Paperclip-Agentfield adapter
- 13. Agentic GRN discovery platform

Some items you listed are not standalone bundles; they are **concretizations inside bundles**. For example, `prepare_docker_runtime_policy`, `check_docker_gpu_access`, `check_kubernetes_context`, and `prepare_remote_compute_profile` belong partly to **Runpod portable runtime base** and partly to **AI Engineer agent/platform dev environment**. I would not make them a fourteenth bundle unless the remote compute story becomes large enough later.